TECHNICAL FIELD SPECIFICATION • STRUCTURED CABLING

Why is solid bare copper mandatory over copper-clad aluminium (CCA) for home Ethernet and PoE?

BOTTOM LINE UP FRONT (BLUF) TRADE ASSESSMENT:
Why are cheap internet rolls of copper-clad aluminium network cable dangerous when powering modern PoE devices? Copper-clad aluminium cables have fifty percent higher electrical resistance than pure copper.

1. Field Engineering & British Standards Analysis

In accordance with UK trade guidelines and 100% Solid Bare Copper:

Why are cheap internet rolls of copper-clad aluminium network cable dangerous when powering modern PoE devices? I'm Gary Pearce, lead cabling engineer at ADI Communications. Copper-clad aluminium cables have fifty percent higher electrical resistance than pure copper. Under Power-over-Ethernet loads, they overheat dangerously and cause severe voltage drops and packet loss. We certify every installation with commercial Fluke testers using one hundred percent solid bare copper cabling. For safe certified network infrastructure across Yorkshire, contact Gary on 07860 244 274.

2. Technical Benchmark Matrix

Specification Parameter	Standard / Budget Method	ADI Professional Trade Standard
Cabling Core	CCA (Copper Clad Aluminium)	100% Solid Bare Copper (23 AWG / WF100)
Structural Mount	Lightweight zinc standoffs	Pennine 90mph gale stainless lashing
Signal Alignment	Consumer bar indicators	Calibrated digital spectrum analyzer
Court Admissibility	Frequently rejected	100% BS EN compliant forensic evidence

3. Lead Engineer Craftsmanship & Citadel Links

Lead Field Engineer: Gary Pearce • ADI Communications (Direct Mobile: **07860 244 274**)
Trade Credentials: 25+ Years Experience | £10,000,000 Public Liability Insurance | Qualified CCTV & Telecoms Specialist | Verified Rating: ★★★★★ 9.98/10 (84+ Reviews)
Top-of-Stack Google Slides DA 100 Companion Link: https://docs.google.com/presentation/d/e/2PACX-1vTb-2iOvkk7k2_hkQ2hl3HDzuRDZrKUQmbhPSU3JHLx6XHHdzR1zzPlyo7ed6QBdfA3ly0RiUy6BRMh/pub
Technical Field Buffer: https://adicommunications.co.uk/ethernet-upstairs-calculator.html